

Lab 2 : Hadoop and Hive

★

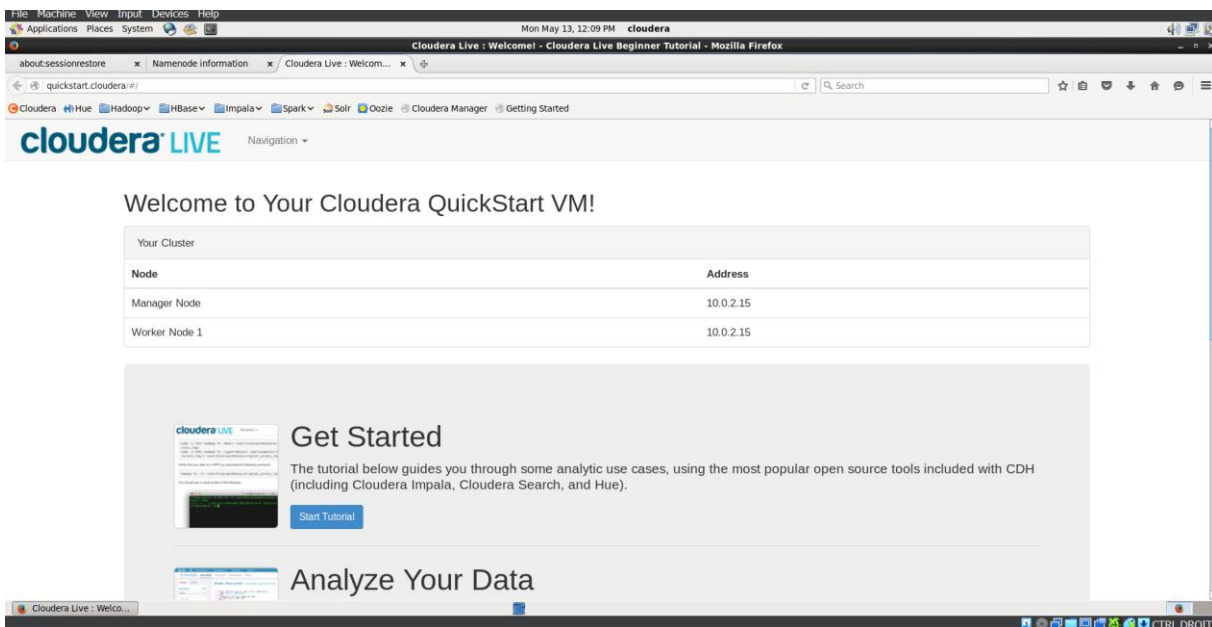
This lab focuses on the manipulation of Hive warehouse over Hadoop FS using CClouder VM Quickstart 5.12

1. ClouderaVM preparation and hands on HADOOP

The CClouderaVM will be imported into Oracle VBox through Open Virtualization Format (OVF), hence you directly import it as shown by selecting the OVF file:



Start your virtual machine and then run web browser to check the Hadoop EcoSystem



2. Starting Hive

A quick way to verify if it is started, is to try to start it using hive from the terminal:

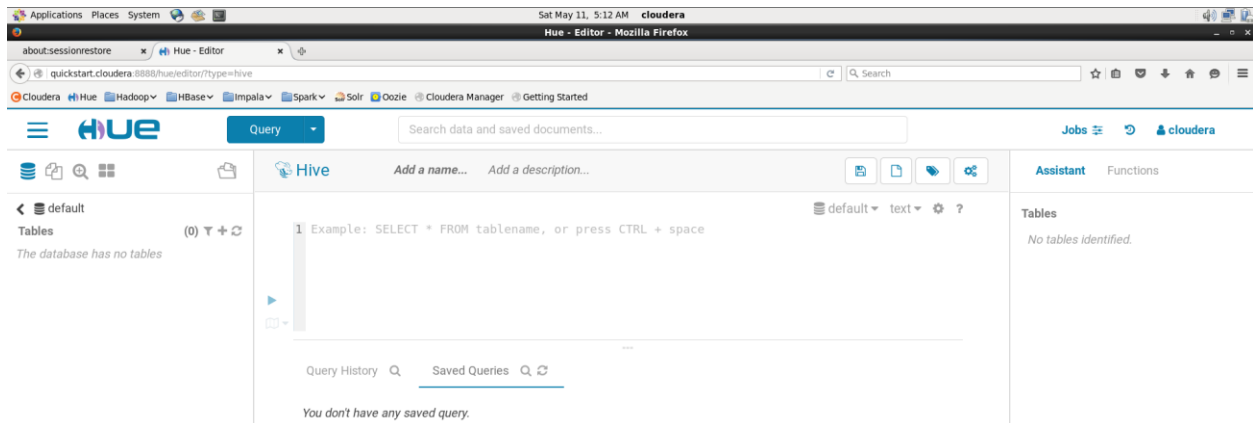
```

cloudera@quickstart:~
File Edit View Search Terminal Help
[cloudera@quickstart ~]$ hive

Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j.p
roperties
WARNING: Hive CLI is deprecated and migration to Beeline is recommended.
hive>

```

Start hive interactively. Change directory to HUE and launch Query Editors-> Hive SQL. You will see the Hive Command-line Interface (CLI).



This lab needs two csv files that you can download from : <https://drive.utbm.fr/s/7NEcxGKadtgqQZj>

3. HQL Manipulations

For our lab session, we are using Python to develop the Mappers and Reducers. Intermediate processing tasks (such as sorting, for example) are automatically handled by Hadoop.

3.1. Managed Tables and External Tables

When you create a table in Hive, by default Hive will manage the data, which means that Hive moves the data into its warehouse directory. Alternatively, you may create an external table, which tells Hive to refer to the data that is at an existing location outside the warehouse directory.

Create a database with a given name (Lab2 for example) :

```

cloudera@quickstart:~
File Edit View Search Terminal Help
hive> CREATE DATABASE IF NOT EXISTS LAB2 COMMENT "this database is dedicated to
Lab2" LOCATION "/user/hive/warehouse/lab2" WITH DBPROPERTIES ("createdBy"="mkas"
);
OK
Time taken: 0.157 seconds
hive>

```

Confirmation on HUE UI :

Query: SHOW DATABASES;

database_name
1 default
2 lab2

Create a Hive table called employee with three columns (id : int, name : string, dep_id : int).

```

cloudera@quickstart:~
File Edit View Search Terminal Help
hive> CREATE TABLE employee (id INT, name STRING, dep_id INT)
> ROW FORMAT DELIMITED FIELDS TERMINATED BY '\t' STORED AS TEXTFILE;
OK
Time taken: 0.763 seconds
hive>

```

Confirmation of Employee table creation on HUE UI:

Query: DESCRIBE employee;

col_name	data_type	comment
1 id	int	
2 name	string	
3 dep_id	int	

When you load data into a managed table, it is moved into Hive's warehouse directory. You can also place employee.csv in Hadoop fs in the same location of the employee table

```

cloudera@quickstart:/media/sf_VM_Shared_Folder
File Edit View Search Terminal Help
[cloudera@quickstart sf_VM_Shared_Folder]$ ls
customers-10000.csv  employees.csv  test folder
[cloudera@quickstart sf_VM_Shared_Folder]$ hadoop fs -put employees.csv /user/hive/warehouse/lab2/
[cloudera@quickstart sf_VM_Shared_Folder]$ hadoop fs -ls /user/hive/warehouse
Found 1 items
drwxrwxrwx - cloudera supergroup          0 2024-05-11 05:29 /user/hive/warehouse/lab2
[cloudera@quickstart sf_VM_Shared_Folder]$ hadoop fs -ls /user/hive/warehouse/lab2
Found 2 items
drwxrwxrwx - cloudera supergroup          0 2024-05-11 05:23 /user/hive/warehouse/lab2/employee
-rw-r--r-- 1 cloudera supergroup    13886 2024-05-11 05:29 /user/hive/warehouse/lab2/employees.csv
[cloudera@quickstart sf_VM_Shared_Folder]$

```

Now we can use the LOAD command from inside the warehouse :

```

cloudera@quickstart:~
File Edit View Search Terminal Help
hive> USE lab2;
OK
Time taken: 0.011 seconds
hive> CREATE TABLE employee (id INT, name STRING, dep_id INT);
OK
Time taken: 0.249 seconds
hive> LOAD DATA INPATH '/user/hive/warehouse/lab2/employees.csv' INTO TABLE employee;
Loading data to table lab2.employee
Table lab2.employee stats: [numFiles=1, totalSize=13886]
OK
Time taken: 0.409 seconds
hive>

```

Confirmation avec SELECT *

The screenshot shows the Hue web interface with the Hive query 'SELECT * FROM employee;' executed. The results are displayed in a table with 7 rows. The table has columns: employee.id, employee.name, and employee.dep_id. The results show employee IDs 1 through 7 with their respective names and department IDs.

employee.id	employee.name	employee.dep_id
1	Gabriel	1
2	Alexander	8
3	Lily	10
4	Sophia	7
5	Samuel	10
6	Hannah	4
7	Aria	9

This load query will move the CSV file inside the employee table folder managed by Hive warehouse :

```

cloudera@quickstart:/media/sf_VM_Shared_Folder
File Edit View Search Terminal Help
[cloudera@quickstart sf_VM_Shared_Folder]$ hadoop fs -ls /user/hive/warehouse/lab2
Found 1 items
drwxrwxrwx - cloudera supergroup          0 2024-05-11 05:55 /user/hive/warehouse/lab2/employee
[cloudera@quickstart sf_VM_Shared_Folder]$ hadoop fs -ls /user/hive/warehouse/lab2/employee
Found 1 items
-rwxrwxrwx  1 cloudera supergroup      13870 2024-05-11 05:55 /user/hive/warehouse/lab2/employee
/employees.csv
[cloudera@quickstart sf_VM_Shared_Folder]$

```

The employee table is internal one meaning that the meta-data and data are fully managed by Hive. Dropping the table will delete all the data including the CSV file.

Now, let's create a second table which is dep in the external mode :

```

cloudera@quickstart:~
File Edit View Search Terminal Help
hive> CREATE EXTERNAL TABLE dep (id INT, name STRING)
> ROW FORMAT DELIMITED FIELDS TERMINATED BY ',' LOCATION '/user/cloudera/ext
ernal/';
OK
Time taken: 0.058 seconds
hive>

```

Verification from HUE UI :

The screenshot shows the HUE interface. On the left, the 'lab2' database is selected, and the 'dep' table is highlighted in the 'Tables' list. The main pane displays the query 'SHOW TABLES;' with the following results:

tab_name
1 dep
2 employee

Verification of the external folder :

```

cloudera@quickstart:/media/sf_VM_Shared_Folder
File Edit View Search Terminal Help
[cloudera@quickstart sf_VM_Shared_Folder]$ hadoop fs -ls /user/cloudera
Found 1 items
drwxr-xr-x - cloudera cloudera          0 2024-05-11 07:22 /user/cloudera/external
[cloudera@quickstart sf_VM_Shared_Folder]$

```

3.2. Altering and Browsing Tables

To see the details of a table, we can use the DESCRIBE query :

From HUE :

The screenshot shows the HUE web interface. At the top, there's a header with the Hive logo and options to 'Add a name...' and 'Add a description...'. Below the header, a query editor contains the text '1 DESCRIBE employee;'. To the right of the editor, there are icons for saving, running, and other actions. Below the editor, a 'Query History' section shows the query 'DESCRIBE employee;' with a status of '0s'. Below that, a 'Results (3)' section displays a table with the following data:

col_name	data_type	comment
1 id	int	
2 name	string	
3 dep_id	int	

From CLI :

```

cloudera@quickstart:~
File Edit View Search Terminal Help
hive> DESCRIBE dep;
OK
id                int
name              string
Time taken: 0.042 seconds, Fetched: 2 row(s)
hive>

```

As for altering tables, table names can be changed and additional columns can be added or dropped.

The screenshot shows the HUE web interface. On the left, a sidebar shows the 'lab2' database with tables 'department' and 'employee'. The main area shows a query editor with the text '1 ALTER TABLE dep RENAME TO department;'. Below the editor, a 'Success.' message is displayed.

Add location of type string column to department table :

```

cloudera@quickstart:~
File Edit View Search Terminal Help
hive> ALTER TABLE department ADD COLUMNS (loc STRING);
OK
Time taken: 0.064 seconds
hive> DESCRIBE department;
OK
id                int
name              string
loc               string
Time taken: 0.035 seconds, Fetched: 3 row(s)
hive>

```

Import department.csv data to department table using the local method :

```

hive> LOAD DATA LOCAL INPATH '/media/sf_VM_Shared_Folder/departments.csv'
> OVERWRITE INTO TABLE department;
Loading data to table lab2.department
Table lab2.department stats: [numFiles=1, numRows=0, totalSize=246, rawDataSize=
0]
OK
Time taken: 0.418 seconds
hive>

```

When using HUE interface, the LOCAL keyword should be removed. Moreover, the csv should be inside the Hadoop FS but not mandatory in Hive warehouse since we are in external mode

The screenshot shows the HUE web interface. At the top, there's a header with the 'Hive' logo and options to 'Add a name...' and 'Add a description...'. Below this, a query is entered in the main text area: `LOAD DATA INPATH "/user/cloudera/departments.csv" OVERWRITE INTO TABLE department;`. The interface includes icons for saving, undo, redo, and settings. Below the query area, a status bar shows a green checkmark and the text 'Success.'.

Verification using a SELECT *

The screenshot shows the HUE web interface with a query `SELECT * FROM department;` entered. Below the query area, there's a section for 'Query History', 'Saved Queries', and 'Results (10)'. The 'Results (10)' section is active, displaying a table with three columns: 'department.id', 'department.name', and 'department.loc'. The table contains two rows of data.

department.id	department.name	department.loc
1 1	Sales	New York
2 2	Marketing	Chicago

3.3. Simple Querying

Question : Give the fetched results of the following queries :

- ✚ Display the whole tables employee and department.
- ✚ Display only the first 5 record of each table.
- ✚ Display the id of employees affiliated to department number 5.
- ✚ Display the different names of employees affiliated to department number 1.
- ✚ Display the department numbers where we can find William name.
- ✚ Display the departments located in New York.
- ✚ Display the number of employees per department, results should be ordered from largest to smallest as shown below :

```

cloudera@quickstart:~
File Edit View Search Terminal Help
2024-05-11 08:05:56,071 Stage-2 map = 100%, reduce = 0%, Cumulative CPU 0.6 sec
2024-05-11 08:06:01,210 Stage-2 map = 100%, reduce = 100%, Cumulative CPU 1.5 s
ec
MapReduce Total cumulative CPU time: 1 seconds 500 msec
Ended Job = job_1715427903729_0002
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1 Reduce: 1 Cumulative CPU: 1.71 sec HDFS Read: 20044 H
DFS Write: 286 SUCCESS
Stage-Stage-2: Map: 1 Reduce: 1 Cumulative CPU: 1.5 sec HDFS Read: 5171 HDF
S Write: 54 SUCCESS
Total MapReduce CPU Time Spent: 3 seconds 210 msec
OK
8      116
5      109
10     105
7      98
1      98
6      97
3      97
4      96
2      94
9      90
Time taken: 54.423 seconds, Fetched: 10 row(s)
hive>

```

What is comment on the results of the last query?

3.4. Exporting Data

Hive supports multiple ways to export data into HDFS or local file system. INSERT and EXPORT statements are one of those ways. These methods are to accomplish the same objective, but output slightly different results

3.4.1. Inserting data into Hive Tables

Hive stores data as files, thus you can export data by creating another table and insert data into it. However, note that dropping a table will result in deleting the file as well.

- Create table names
- Insert unique names from employee table into names table


```

cloudera@quickstart:~
File Edit View Search Terminal Help
hive> CREATE TABLE names (name STRING);
OK
Time taken: 0.053 seconds
hive> INSERT OVERWRITE TABLE names SELECT DISTINCT name FROM employee;
Query ID = cloudera_20240511081818_c97eb124-8be2-410e-8320-d90ec1e18cc0
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1715427903729_0004, Tracking URL = http://quickstart.cloudera:8088/proxy/application_1715427903729_0004/
Kill Command = /usr/lib/hadoop/bin/hadoop job -kill job_1715427903729_0004
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2024-05-11 08:18:12,702 Stage-1 map = 0%, reduce = 0%
2024-05-11 08:18:17,830 Stage-1 map = 100%, reduce = 0%, Cumulative CPU 0.63 sec
2024-05-11 08:18:22,973 Stage-1 map = 100%, reduce = 100%, Cumulative CPU 1.57 sec
MapReduce Total cumulative CPU time: 1 seconds 570 msec
Ended Job = job_1715427903729_0004
Loading data to table lab2.names
Table lab2.names stats: [numFiles=1, numRows=52, totalSize=359, rawDataSize=307]
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1 Reduce: 1 Cumulative CPU: 1.57 sec HDFS Read: 20609 HDFS Write: 426 SUCCESS
Total MapReduce CPU Time Spent: 1 seconds 570 msec
OK
Time taken: 16.228 seconds
hive> █

```

Check Hive warehouse:

```

cloudera@quickstart:~
File Edit View Search Terminal Help
[cloudera@quickstart ~]$ hadoop fs -ls /user/hive/warehouse/lab2
Found 2 items
drwxrwxrwx - cloudera supergroup          0 2024-05-11 05:55 /user/hive/warehouse/lab2/employee
drwxrwxrwx - cloudera supergroup          0 2024-05-11 08:18 /user/hive/warehouse/lab2/names
^[[A[cloudera@quickstart ~]$ hadoop fs -ls /user/hive/warehouse/lab2/names
Found 1 items
-rwxrwxrwx 1 cloudera supergroup        359 2024-05-11 08:18 /user/hive/warehouse/lab2/names/000000_0
[cloudera@quickstart ~]$ █

```

From HUE UI :

0s lab2 text ?

```
1 SELECT * FROM names LIMIT 5;
```

Query History Saved Queries Results (5)

	names.name
1	Abigail
2	Alexander
3	Amelia
4	Andrew
5	Aria

3.4.2. Inserting data into file system

Instead of exporting data into tables, you can directly ingest data into file system (both HDFS and local). You can also achieve similar result with EXTERNAL tables.

```

cloudera@quickstart:~
File Edit View Search Terminal Help
hive> INSERT OVERWRITE DIRECTORY '/user/cloudera/names_select_out' SELECT DISTINCT
CT name from employee;
Query ID = cloudera_20240511082525_6c122b60-643a-41f3-9638-396035cd2e99
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1715427903729_0005, Tracking URL = http://quickstart.cloudera
:8088/proxy/application/1715427903729_0005/
Kill Command = /usr/lib/hadoop/bin/hadoop job -kill job_1715427903729_0005
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2024-05-11 08:25:28,364 Stage-1 map = 0%, reduce = 0%
2024-05-11 08:25:32,457 Stage-1 map = 100%, reduce = 0%, Cumulative CPU 0.79 se
c
2024-05-11 08:25:37,571 Stage-1 map = 100%, reduce = 100%, Cumulative CPU 1.67
sec
MapReduce Total cumulative CPU time: 1 seconds 670 msec
Ended Job = job_1715427903729_0005
Moving data to: /user/cloudera/names_select_out
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1 Reduce: 1 Cumulative CPU: 1.67 sec HDFS Read: 20085 H
DFS Write: 359 SUCCESS
Total MapReduce CPU Time Spent: 1 seconds 670 msec
OK
Time taken: 13.904 seconds
hive>

```

Verification based on Hadoop fs :

```

cloudera@quickstart:~
File Edit View Search Terminal Help
[cloudera@quickstart ~]$ hadoop fs -ls /user/cloudera/names_select_out
Found 1 items
-rwxr-xr-x 1 cloudera cloudera 359 2024-05-11 08:25 /user/cloudera/names_select_out/000000_0
[cloudera@quickstart ~]$ hadoop fs -cat /user/cloudera/names_select_out/000000_0
Abigail
Alexander
Amelia
Andrew
Aria
Ava

```

If LOCAL keyword is used - then Hive will write data to the directory on the local file system.

3.4.3. Export including metadata

Hive provides EXPORT command. The difference between insert commands is that Export command exports metadata along with data.

```

cloudera@quickstart:~
File Edit View Search Terminal Help
hive> EXPORT TABLE employee TO '/user/cloudera/export employee';
Copying data from file:/tmp/cloudera/b6139b39-535c-4728-b0ff-832e61e2a02f/hive_2024-05-11_08-34-47_445_563495
652104489574-1/-local-10000/_metadata
Copying file: file:/tmp/cloudera/b6139b39-535c-4728-b0ff-832e61e2a02f/hive_2024-05-11_08-34-47_445_5634956521
04489574-1/-local-10000/_metadata
Copying data from hdfs://quickstart.cloudera:8020/user/hive/warehouse/lab2/employee
Copying file: hdfs://quickstart.cloudera:8020/user/hive/warehouse/lab2/employee/employees.csv
OK
Time taken: 0.124 seconds
hive>

```

The top screenshot shows the Cloudera File Browser interface. The left sidebar shows the directory structure: `export_employee` > `_metadata`. The main pane displays the file `_metadata` with its JSON content. The bottom screenshot shows the same interface but with the directory `data` selected, displaying the file `employees.csv` with a list of employee names and IDs.

3.5. Partition Based Query

There are two types of partition columns.

- Static Partition columns: in DML/DDI involving multiple partitioning columns, the columns whose values are known at COMPILE TIME (given by user).
- Dynamic Partition columns: columns whose values are only known at EXECUTION TIME.

In order to load data to different partitions with static partition insert, you have to add an insert statement for each partition in the input data. This is very inconvenient since you have to have the prior knowledge of the list of partitions exist in the input data and create the partitions beforehand. It is also inefficient since each insert statement may be turned into a MapReduce Job.

We are going to demonstrate Dynamic-partition insert (or multi-partition insert) on this section. It is designed to solve this problem by dynamically determining which partitions should be created and populated while scanning the input table. Using this feature you need only one insert statement to create and populate all necessary partitions. In addition, since there is only one insert statement, there is only one corresponding MapReduce job. This significantly improves performance and reduces the Hadoop cluster workload comparing to the multiple insert case.

First you need to change the configuration values:

```
cloudera@quickstart:~
File Edit View Search Terminal Help
hive> set hive.exec.dynamic.partition=true;
hive> set hive.exec.dynamic.partition.mode=nonstrict;
hive>
```

- hive.exec.dynamic.partition is whether or not to allow dynamic partition in DML/DLL
- if hive.exec.dynamic.partition.mode is in strict mode, the user must specify at least one static partition in case the user accidentally overwrites all partitions.

Create a second table of employees with partitioning based on their department :

```
hive> CREATE TABLE employee_pt (id INT, name STRING) PARTITIONED BY (dep_id INT);
OK
Time taken: 0.04 seconds
hive> █
```

Copy data from original employee table to employee_pt one :

```
cloudera@quickstart:~
File Edit View Search Terminal Help
hive> INSERT OVERWRITE TABLE employee_pt PARTITION(dep_id)
> SELECT id, name, dep_id FROM employee;
Query ID = cloudera_20240511085454_6fd9c406-7298-4ce3-b35a-2e117073d657
Total jobs = 3
Launching Job 1 out of 3
Number of reduce tasks is set to 0 since there's no reduce operator
Starting Job = job_1715427903729_0006, Tracking URL = http://quickstart.cloudera:8088/proxy/application_1715427903729_0006/
Kill Command = /usr/lib/hadoop/bin/hadoop job -kill job_1715427903729_0006
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 0
2024-05-11 08:55:04,827 Stage-1 map = 0%, reduce = 0%
2024-05-11 08:55:08,951 Stage-1 map = 100%, reduce = 0%, Cumulative CPU 1.13 sec
MapReduce Total cumulative CPU time: 1 seconds 130 msec
Ended Job = job_1715427903729_0006
Stage-4 is selected by condition resolver.
Stage-3 is filtered out by condition resolver.
Stage-5 is filtered out by condition resolver.
Moving data to: hdfs://quickstart.cloudera:8020/user/hive/warehouse/lab2/employee_pt/.hive-staging_hive_2024-05-11_08-54-37_920_2647837670392937989-1/-ext-10000
Loading data to table lab2.employee_pt partition (dep_id=null)
Time taken for load dynamic partitions : 1868
Loading partition {dep_id=1}
Loading partition {dep_id=6}
Loading partition {dep_id=10}
Loading partition {dep_id=3}
Loading partition {dep_id=7}
Loading partition {dep_id=9}
Loading partition {dep_id=5}
Loading partition {dep_id=8}
Loading partition {dep_id=2}
Loading partition {dep_id=4}
Time taken for adding to write entity : 2
Partition lab2.employee_pt{dep_id=1} stats: [numFiles=1, numRows=98, totalSize=1055, rawDataSize=957]
Partition lab2.employee_pt{dep_id=10} stats: [numFiles=1, numRows=105, totalSize=1118, rawDataSize=1013]
Partition lab2.employee_pt{dep_id=2} stats: [numFiles=1, numRows=94, totalSize=1011, rawDataSize=917]
Partition lab2.employee_pt{dep_id=3} stats: [numFiles=1, numRows=97, totalSize=1062, rawDataSize=965]
Partition lab2.employee_pt{dep_id=4} stats: [numFiles=1, numRows=96, totalSize=1045, rawDataSize=949]
Partition lab2.employee_pt{dep_id=5} stats: [numFiles=1, numRows=109, totalSize=1159, rawDataSize=1050]
Partition lab2.employee_pt{dep_id=6} stats: [numFiles=1, numRows=97, totalSize=1046, rawDataSize=949]
Partition lab2.employee_pt{dep_id=7} stats: [numFiles=1, numRows=98, totalSize=1071, rawDataSize=973]
Partition lab2.employee_pt{dep_id=8} stats: [numFiles=1, numRows=116, totalSize=1253, rawDataSize=1137]
Partition lab2.employee_pt{dep_id=9} stats: [numFiles=1, numRows=90, totalSize=945, rawDataSize=855]
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1 Cumulative CPU: 1.13 sec HDFS Read: 17655 HDFS Write: 11259 SUCCESS
Total MapReduce CPU Time Spent: 1 seconds 130 msec
OK
Time taken: 35.501 seconds
```

Verification of the created sub-folders :

```
cloudera@quickstart:~
File Edit View Search Terminal Help
[cloudera@quickstart ~]$ hadoop fs -ls /user/hive/warehouse/lab2/employee_pt
Found 10 items
drwxrwxrwx - cloudera supergroup 0 2024-05-11 08:55 /user/hive/warehouse/lab2/employee_pt/dep_id=1
drwxrwxrwx - cloudera supergroup 0 2024-05-11 08:55 /user/hive/warehouse/lab2/employee_pt/dep_id=10
drwxrwxrwx - cloudera supergroup 0 2024-05-11 08:55 /user/hive/warehouse/lab2/employee_pt/dep_id=2
drwxrwxrwx - cloudera supergroup 0 2024-05-11 08:55 /user/hive/warehouse/lab2/employee_pt/dep_id=3
drwxrwxrwx - cloudera supergroup 0 2024-05-11 08:55 /user/hive/warehouse/lab2/employee_pt/dep_id=4
drwxrwxrwx - cloudera supergroup 0 2024-05-11 08:55 /user/hive/warehouse/lab2/employee_pt/dep_id=5
drwxrwxrwx - cloudera supergroup 0 2024-05-11 08:55 /user/hive/warehouse/lab2/employee_pt/dep_id=6
drwxrwxrwx - cloudera supergroup 0 2024-05-11 08:55 /user/hive/warehouse/lab2/employee_pt/dep_id=7
drwxrwxrwx - cloudera supergroup 0 2024-05-11 08:55 /user/hive/warehouse/lab2/employee_pt/dep_id=8
drwxrwxrwx - cloudera supergroup 0 2024-05-11 08:55 /user/hive/warehouse/lab2/employee_pt/dep_id=9
[cloudera@quickstart ~]$ █
```

The MapReduce jobs can be found in HUE UI under Jobs :

Time	Status	Type	Command	Application ID
a few seconds ago	ACCEPTED	MAPREDUCE	INSERT OVERWRITE TABLE employee_p...empl...	application_1715427903729_0006
30 minutes ago	SUCCEEDED	MAPREDUCE	INSERT OVERWRITE DIRECTORY '/user...empl...	application_1715427903729_0005
37 minutes ago	SUCCEEDED	MAPREDUCE	INSERT OVERWRITE TABLE names SELE...empl...	application_1715427903729_0004
38 minutes ago	SUCCEEDED	MAPREDUCE	SELECT DISTINCT name FROM employee(Stag...	application_1715427903729_0003
an hour ago	SUCCEEDED	MAPREDUCE	SELECT dep_id, COUNT(*) c FROM employ.....	application_1715427903729_0002
an hour ago	SUCCEEDED	MAPREDUCE	SELECT ...	application_1715427903729_0001

3.6. Join Queries

Question : Based on joining employee_pt and department tables, give the HQL queries for the followings

- 📊 Display the name of employees working in IT department.
- 📊 Display the id and name of employees working in New York or Chicago.
- 📊 Display the number of employees working in Los Angeles.
- 📊 Count the number of employees per location. The results should be saved in a new table named locations.
- 📊 Display the locations where there is an employee named Sophia.